



F-35 fighter data stolen

Sensitive data about Australia's F-35 stealth fighter and P-8 surveillance aircraft programmes were stolen in a defence hack <http://dgit.in/F35Hack>



Millimetre wave to power VR?

Research from Columbia Engineering indicates that millimetre wave frequency could make truly wireless VR a reality <http://dgit.in/MilliVR>

Intel Core i7 8700K

Scaling up with six cores

Intel's Coffee Lake comes close on the heels of its own Basin Falls platform. And it goes without saying that AMD's push for more cores in the CPU segment has led to one of the key changes that we see with the Intel Core i7 8700K – bye, bye incremental upgrades (at least we hope so). With Coffee Lake, Intel is bringing two additional cores across their CPU stack. This means, you'll now get quad-core Intel Core i3 CPUs and hexa-core Intel Core i5 and Intel Core i7 processors.

Before we get into the performance aspects, let's take a look at the new Coffee Lake platform.

Intel Coffee Lake Platform

The Coffee Lake platform is realised with Coffee Lake CPUs and Intel Z370 chipset based motherboards. Unlike all previous launches, Intel has reduced the number of chipset variants for Coffee Lake at launch. We'll only see Z370 based motherboards for now. But if leaks are to be believed scaled down boards around chipsets such as the H370, H310 and B360 chipsets will launch in Q1 2018. Intel will continue support for Optane Memory on the Series-300 chipsets. Optane offers performance which is a hybrid between existing SSDs and RAM technologies.

Intel Core i7-8700K performance

We'll be comparing the Intel Core i7-8700K and Intel Core i5-8400 against the existing Intel Core i7-7700K, Core i9-7900X, Core i7-7740X, AMD Ryzen 7 1800X, 1700X, 1700 and Ryzen 5 1600X. Since our Test Rig has changed post Skylake, we'll need to retest with our newer components. For now, since the Core i7-8700K is an unlocked SKU (which is in a way meant for overclocking), we had to push it a little. We started off directly with the built-in profiles on the GIGABYTE

Z370 AORUS Gaming 7 and the CPU easily scaled to 5.0 GHz. We chose to use air-cooling for the session. Nothing fancy, not even an AIO. The Noctua NH-D14 was more than sufficient for what we were doing. Getting the 8700K to 5.1 GHz needed a bit more Vcore. We found 1.41 to be sufficient for running the entire set of benchmarks at 5.1 GHz. However, 5.2 GHz wasn't that easy. Bumping the Vcore to 1.43 was more than enough to get the PC to boot.

However, it didn't help with the benchmarks and we started noticing some really high temps. Most often, we'd find the CPU throttling or the benchmark crashing. So that's where we decided to stop with the Intel Core i7-8700K till we get our hands on a better cooling solution.

The key takeaway from this is that running mainstream CPUs at 5-5.1 GHz on air-cooling is easy. These tests were performed on an open bench with the ambient temperature being 24 degrees Celsius. The low point was obviously the CPU set to stock frequency but the RAM set to 2133 MHz which is lower than the default 2667 MHz.

The percentile gain between the high-point and the low-point was 15.95% which is more than what you get between CPU generations. This was discerned from the multi-threaded runs on Cinebench R15. With single-threaded runs, the performance gain shrunk to 10% which is, again, a decent gain.

Verdict

Finally, we have a six-core Intel CPU in the mainstream segment. And the good thing about the Intel Core i7-8700K is that it's a 6-core processor that outdoes AMD's 8-core



Price
₹27,000



FEATURES.....	75
PERFORMANCE.....	90
VALUE.....	75
DESIGN.....	75

Ryzen 7 1800X in quite a few benchmarks. Given that its US pricing is lower than the 1800X's pricing, Intel has officially given rise to a better CPU. Intel's weakness was the lower core-count in the mainstream segment, which AMD clearly took advantage of.

AMD's monumental IPC increment was still no match for Intel's capabilities. So despite the apparent lack of IPC improvement in Coffee Lake vs Kabylake, we still end up getting a better performance overall. There are rendering scenarios which are heavily multi-threaded where the Ryzen 7 1800X continues to shine but the gap has been reduced greatly. Gaming is an area where AMD still needs a lot of optimisation but Coffee Lake doesn't. We tried overclocking the CPU on the GIGABYTE Z370 AORUS Gaming 7 and managed to get it stable at 5.1 GHz on air-cooling. We even took it up to 5.2 GHz but not all benchmarks could run without crashes. All that remains is for the motherboard manufacturers to release a budget Z370 board and you have your next upgrade ready.

—Mithun Mohandas

SPECIFICATIONS

CLOCK SPEED: 3.7 GHz | TURBO FREQUENCY: 4.7 GHz | CORES/ THREADS: 6/12 | LITHOGRAPHY: 14nm | TDP: 95W | PCIE LANES: 16 | PCIE VERSION: 3.0 | MEMORY SUPPORT: DDR4.

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